

SCHOOL MANAGEMENT PLATFORM

Complete Brainstorming & Vision Document

India-First · AI-Native · 100% Customizable · SaaS

1. Product Vision

The platform is a fully customizable, AI-native school management operating system built for the Indian education market. Every school in India has a unique management structure — different hierarchies, workflows, fee structures, academic calendars, and reporting needs. Today's solutions force schools into rigid templates.

This platform inverts that model: a powerful default system ships out of the box, and everything beyond that — roles, workflows, modules, UI, data structures — is configured by the school itself through a Figma-like drag-and-drop canvas or by the built-in AI agent.

Core Promise

Buy a subscription → Get one super admin account → Configure everything from roles to workflows to fees to salaries — visually, without writing a single line of code. Teach with built-in digital books, assignments, and a full learning calendar. Deploy on school tablets with one click. The backend adapts automatically as the frontend changes.

2. Product Name

Working name: Vidyut (विद्युत् — Sanskrit for electric current / divine energy). Short, 6 characters, easy to type in code, works in every Indian language. Final name to be decided later — rename is a one-click global find-replace across the entire codebase.

Confirmed Available Domains

gurukrama.com is confirmed available. Other strong candidates verified: Samagraha (samagra = complete/whole, resonates with NEP 2020), Bodhasetu (bodha + setu = bridge of awakening, Ramayana reference), Vishvavidya (vishva + vidya = universal knowledge, Bhagavad Gita Chapter 11), Prashthan (departure/beginning of a journey, Prasthanatrayi reference). Nalanda and Takshashila have .com taken but .io/.ai variants likely available. Name decision deferred — product ships as Vidyut until chosen.

3. Target Market

3.1 Primary Buyer Personas

Persona	Description
School Owner / Trust	Private school owners, management trusts running single or multi-branch schools. Decision-makers who sign cheques and care about ROI, compliance, and reputation.
Principal / Headmaster	Day-to-day operational authority. Wants efficiency, visibility, and tools that reduce administrative burden on staff.
Government / Municipal Schools	Larger procurement cycles but massive scale. May require custom compliance modules (RTE, NEP alignment, state board requirements).

3.2 School Size Coverage

The platform targets all school sizes from day one, with pricing tiers that reflect scale:

- Small schools (under 500 students) — simplified onboarding, core modules only, lowest tier pricing
- Medium schools (500–2000 students) — full module suite, multi-department management
- Large schools (2000+ students) — enterprise features, multi-branch support, advanced analytics, dedicated onboarding support

3.3 Geography

India-first launch. Initial focus on Tier 1 and Tier 2 cities where private school density is highest (Hyderabad, Bangalore, Chennai, Pune, Mumbai, Delhi NCR). Expand to Tier 3 cities in Year 2. International expansion (Southeast Asia, Middle East Indian diaspora schools) in Year 3+.

4. Business Model

4.1 Subscription Tiers (SaaS)

Tier	Inclusions
Starter	Small schools up to 500 students. Core modules: admissions, attendance, timetable, fees, basic reports. Monthly or annual billing.
Growth	Schools up to 2000 students. All Starter features + HR/payroll, advanced fee management, parent app, transport, library, education content module.

Enterprise	Unlimited students + multi-branch + full learning platform + custom module development + dedicated AI agent training + priority support + onboarding by company team.
Vidyut Learn (Add-on)	Education content tier — available as add-on to any subscription. Unlocks digital books, teacher content creation, assignment builder, full learning calendar, student progress tracking, and AI-powered personalised learning paths.
Vidyut Device (Add-on)	Tablet deployment tier — for schools with tablets or partnered devices. Offline-capable learning app, content sync, device management dashboard, usage analytics, parental controls. Hardware partnership programme available.

4.2 Additional Revenue Streams

- **Company-assisted setup for schools that prefer white-glove configuration (one-time fee).** Onboarding & Setup Service
- **Schools can request bespoke modules beyond the drag-and-drop builder for complex needs.** Custom Module Development
- **Advanced AI features (bulk data analysis, predictive insights, auto-report generation) as add-ons.** AI Agent Premium Usage
- **Staff training programs for school administrators.** Training & Certification

5. Core Architecture Concept

The most technically ambitious aspect of this product is the live schema adaptation — when a user changes the frontend (adds a field, creates a new workflow, defines a new role), the backend database structure updates automatically. This is the backbone of the '100% customizable' promise.

5.1 Three-Layer Customization System

Layer	What It Does
Layer 1: Role Engine	Drag-and-drop canvas to define roles, permissions, hierarchies. Create, edit, clone, delete roles. Assign permissions at module, page, field, and action level.
Layer 2: Workflow Builder	Figma-like canvas to design workflows (admissions flow, fee collection flow, leave approval chain, etc.). Each node in the workflow triggers backend logic.
Layer 3: Schema Engine	When a new field, form, or workflow is created in the UI, the system auto-provisions database tables, columns, and relationships. No manual migration required.

5.2 Access Control Model

One super admin account per school is issued at purchase. The super admin has unlimited access including creating other admin users, roles, and sub-admins. All other roles and their permissions are defined by the super admin or the AI agent.

Key Differentiator

No other Indian school ERP dynamically adapts its database schema based on UI changes. This is what makes the platform truly school-agnostic — not just configurable at the surface level, but structurally adaptive at the data layer.

6. Core Modules (Default Out-of-the-Box)

These modules ship with every subscription as the default school management foundation. Schools can extend, modify, or disable any module.

6.1 Academic Management

- — **Application forms, document collection, entrance test management, offer letters, enrollment confirmation.** Admissions & Enrollment
- — **Complete student profiles, academic history, behavior records, medical info, family details.** Student Records
- — **Drag-and-drop period scheduling, teacher allocation, room management, clash detection.** Timetable & Scheduling
- — **Daily and period-wise attendance, biometric integration, parent notifications, reports.** Attendance
- — **Exam scheduling, marks entry, grade computation, report card generation, rank lists.** Examination & Results
- — **Lesson plan tracking, syllabus completion, learning outcomes mapping.** Curriculum Management

6.2 Finance Management

- — **Fee structure builder (by class/category/installment), online & offline collection, receipts, defaulter tracking.** Fee Management
- — **Annual budget planning, department-wise budget allocation, expense tracking, variance analysis.** Budgeting
- — **Staff salary computation, allowances, deductions, PF/ESI, TDS, payslip generation, bank transfer files.** Payroll & Salaries
- — **Income/expense ledger, day book, bank reconciliation, financial statements.** Accounts & Ledger
- — **Vendor management, purchase orders, invoice tracking, payment approvals.** Vendor & Procurement

6.3 Human Resources

- — **Employee records, qualifications, experience, documents, contracts.** Staff Profiles
- — **Job posting, application tracking, interview scheduling, offer management.** Recruitment
- — **Leave types, application workflows, approval chains, balance tracking.** Leave Management
- — **Appraisal cycles, goal setting, review forms, increment workflows.** Performance Management
- — **Training calendar, attendance, certifications, skill tracking.** Training & Development

6.4 Operations Management

- — **Route management, vehicle tracking, driver records, student bus assignment, GPS integration hook.** Transport
- — **Room allocation, fee management, visitor log, warden dashboard.** Hostel
- — **Book catalog, issue/return, fine management, digital resources.** Library
- — **Asset register, maintenance scheduling, depreciation tracking, consumables stock.** Inventory & Assets
- — **Menu management, daily orders, billing, nutrition tracking.** Canteen
- — **Event calendar, participation tracking, co-curricular records.** Events & Activities

6.5 Communication & Engagement

- — **Real-time notifications, fee payment, attendance view, report access, teacher messaging.** Parent Portal & App
- — **Homework, results, timetable, announcements, digital content access.** Student Portal
- — **Staff announcements, department circulars, noticeboard, messaging.** Internal Communication
- — **Automated alerts for fees, attendance, results, events.** SMS & WhatsApp Integration

6.6 Compliance & Reporting

- — **UDISE data, RTE compliance, state board reporting, inspection readiness.** Government Compliance
- — **School-wide KPIs, attendance trends, fee collection analytics, academic performance overview.** Analytics Dashboard
- — **Drag-and-drop report builder to create any report from any data in the system.** Custom Report Builder

6.7 Education Content & Learning Platform (Vidyut Learn)

A complete digital learning layer built into the platform. Teachers create, publish, and assign content. Students consume it on web or tablet. This is not a separate EdTech app — it is fully integrated with attendance, timetable, results, and the AI agent.

- — **School can upload or purchase digital textbooks, workbooks, and reference materials. Organised by class, subject, and chapter. Students access from web or tablet app.** Digital Book Library
- — **Teachers create lessons using a rich content editor: text, images, video embeds, audio, PDFs, interactive quizzes. No external tool needed. All content lives inside the platform.** Teacher Content Studio
- — **Teachers create assignments with due dates, marks, file submission, MCQ, short answer, or essay formats. Auto-grading for MCQ. Manual grading with rubric for essays. Bulk feedback via AI.** Assignment Builder
- — **Integrated calendar showing: class schedule, assignment due dates, exam dates, events, holidays, and teacher-created learning milestones. Synced across parent, student, and teacher views.** Full Learning Calendar
- — **Built-in video class scheduling. Teachers schedule a live session from the timetable. Students join from their portal or tablet. Recording stored automatically in the lesson library.** Live Classes
- — **Per-student view: assignments completed, scores, time spent on content, learning streak, weak topics identified by AI. Visible to teacher, parent, and student.** Student Progress Dashboard
- — **AI analyses student performance data and suggests a personalised revision plan: 'Riya is weak in Algebra — here are 5 exercises to complete before the exam.'** AI-Powered Learning Paths
- — **Students can discuss assignments in a moderated classroom discussion board. Teacher can pose discussion questions. Fully monitored, no direct messaging between students.** Peer Collaboration
- — **Schools can optionally purchase curated content packs: CBSE, ICSE, or State Board aligned digital books and lesson plans from verified publishers. Platform takes commission.** Content Marketplace

6.8 Device & Tablet Tier (Vidyut Device)

For schools that have tablets or partner with our hardware programme, the Device tier deploys the full learning platform directly onto student devices with offline capability, device management, and usage analytics.

- — **Students download assigned content to their tablet. Lessons, books, assignments, and quizzes work fully offline. Syncs to server when Wi-Fi is available. Critical for schools in areas with poor connectivity.** Offline-First Tablet App
- — **School admin sees all registered devices: battery level, last sync time, content downloaded, app version. Remotely push content updates to all devices in one click.** Device Management Dashboard

- — **Teacher publishes a new assignment → all registered tablets receive it automatically next time they connect to school Wi-Fi. Zero manual distribution.** Content Sync Engine
- — **Track time-on-task per student per device. Which lessons were read, which were skipped, how long each quiz took. Data feeds into the Student Progress Dashboard.** Usage Analytics
- — **Parents can see what their child accessed on the device. Set usage time limits. Content is school-curated only — no open internet access in the default device profile.** Parental Controls
- — **Schools that don't have tablets can apply for our partner programme. We connect them with vetted hardware vendors (Acer, Lenovo, local Indian OEMs) for bulk pricing. School gets pre-configured tablets with Vidyut Device pre-installed.** Hardware Partnership Programme
- — **A single tablet can be shared between multiple students (common in government schools) with individual logins and progress tracked separately per student.** Multi-Device Profiles
- — **Track warranty, repair history, insurance claims, and depreciation for every device in the school inventory. Integrated with the main Inventory & Assets module.** Device Lifecycle Management

Why Device Tier Matters

India has 260 million school students. Government's PM SHRI and Digital India programmes are pushing tablet deployment in schools. By offering a turnkey device management + learning platform tier, Vidyut becomes the infrastructure layer for digital learning in Indian schools — not just a management tool. This is the path to 10x scale.

7. Built-In AI Agent

The AI agent is not a chatbot bolted onto the software — it is a core interface layer that can operate the entire system on behalf of any user. It is the fastest path to using and customizing the platform.

7.1 Interaction Modes

- — **Sidebar assistant available on every screen. Context-aware based on the current module.** Chat Interface
- — **Speak to the agent to execute tasks (add a student, generate a fee report, change a timetable slot). Supports Indian English and regional accent handling.** Voice Interface

7.2 What the AI Agent Can Do

Capability	Description
Setup & Configuration	Guide the super admin through initial school setup. Ask questions, configure roles, build default workflows, set up fee structures —

	conversationally.
Role & Permission Design	A principal describes what the accountant should and shouldn't access — the AI agent translates this into a complete role configuration on the canvas.
Workflow Creation	Describe a process in plain English or Hindi — 'when a student applies, the coordinator reviews, then principal approves, then fee is generated' — and the AI builds the workflow.
Data Operations	Query any data conversationally. 'Show me all students with fees due above ₹10,000 in Class 10.' 'Generate April payroll for non-teaching staff.'
Teacher Assistance	Help teachers fill attendance, create exam schedules, generate progress reports, draft parent communication.
Student Assistance	Help students check timetables, fee status, exam schedules, submit leave requests.
Anomaly Detection	Proactively flag issues — unusual expense spikes, attendance drops, fee default patterns — and suggest actions.
Onboarding Guide	Walk any new user through their specific role and the features available to them.

8. Drag-and-Drop Customization Canvas

The customization canvas is the heart of the platform's flexibility. Inspired by Figma's spatial canvas model, it gives admins a visual workspace to build and modify the school's operational structure without coding.

8.1 Role Builder Canvas

- Create roles by dragging role blocks onto the canvas
- Connect roles in a hierarchy using arrows (Principal → Vice Principal → HOD → Teacher)
- Click any role block to assign module permissions, page access, field-level visibility, and action rights
- Clone existing roles and modify — saves time for similar roles
- Delete roles with automatic reassignment flow for existing users in that role

8.2 Workflow Builder Canvas

- Drag trigger nodes (form submission, date event, payment received) onto the canvas
- Connect to action nodes (send notification, create record, assign task, require approval)
- Add condition branches (if fees paid → generate receipt; if not → send reminder)
- Every workflow saved auto-generates the backend logic and any required database structures

8.3 Module & Page Builder

- Add new pages/screens to any module by dragging field components (text, number, date, dropdown, file upload, signature)
- Every new field added to a page auto-provisions a column in the relevant database table
- New tables are created automatically when a completely new data entity is defined
- Forms, lists, and detail views are all configurable without code

Technical Note (Backend Sync)

This is the most complex engineering challenge: a meta-schema system that tracks the logical data model and syncs it with the physical database. Every UI change triggers a schema migration pipeline that is transactional, versioned, and rollback-safe. This must be designed carefully to prevent data loss during schema changes.

9. Technology Directions (To Be Decided)

No final decisions yet — these are directions to explore and finalize in the architecture phase.

Layer	Directions
Frontend	React or Next.js. Custom UI component library (no third-party component kits — all built in-house as stated). Canvas builder likely needs a graph/node library (React Flow or custom).
Backend	Node.js / Python FastAPI. Multi-tenant architecture — one deployment, isolated data per school.
Database	PostgreSQL as primary store with dynamic schema management. Possibly a meta-schema layer on top. Consider schema-per-tenant for isolation.
AI Agent	Claude API (Anthropic) for reasoning. Voice: Whisper or Sarvam AI for Indian English STT. TTS for voice output.
Real-time	WebSockets for live dashboard updates, attendance, notifications.
Auth	Role-based access control (RBAC) with dynamic role definitions from the canvas.
Mobile	React Native or Progressive Web App (PWA) for parent and student portals.
Infrastructure	AWS / GCP. Multi-region for reliability. S3-compatible storage for documents.

10. Competitive Landscape in India

Competitor	Gap We Exploit
Fedena	Open-source school ERP. Popular but dated UI, limited customization, no AI.
Entab (CampusCare)	Widely used in Delhi/NCR private schools. Feature-rich but rigid, expensive custom changes.
Teachmint	Modern UI, teacher-centric. Strong in content delivery but weak in full school operations management.
Skooltek	Strong in South India. Good fee management but limited workflow customization.
MyClassBoard	Comprehensive modules but complex setup, not drag-and-drop.
ERP Next (Education)	Highly customizable but requires technical expertise — not school-admin friendly.

Our Differentiator vs All Competitors

No competitor offers: (1) true drag-and-drop backend-synced customization, (2) a built-in AI agent that can configure and operate the entire system, (3) dynamic schema adaptation when workflows change. We are not building a better school ERP — we are building the OS that schools run on.

11. Backend Architecture — Cloud & Data Layer

Every school gets a fully isolated, fully automated cloud environment. Schools never see or touch cloud infrastructure — they interact only with the platform UI. All provisioning, scaling, backups, and migrations are managed entirely by the platform.

11.1 Two-Plane Architecture

Control Plane + Data Plane

The system is split into two planes. The Control Plane is our platform's brain — it knows about every school, every subscription, every provisioned resource. The Data Plane is each school's isolated cloud environment — their own database, storage, and config. These two planes never mix school data with each other.

Plane	What It Contains
Control Plane	Our central master system. Runs on a dedicated cloud project we own. Contains: master school registry, billing & subscription engine, provisioning service, monitoring dashboard, admin portal for our team. Schools have zero access to this layer.
Data Plane (per school)	Each school's own isolated environment. Contains: their

	PostgreSQL database, Redis instance, file storage bucket, environment config. Created automatically on subscription. Destroyed or archived on cancellation.
--	---

11.2 Per-School Isolated Database

Each school gets a completely separate PostgreSQL database instance — not a separate table, not a separate schema within a shared database, but a fully isolated database. This is the right choice for this product for these reasons:

- — **a bug in one school's queries can never leak another school's data. Zero cross-contamination risk.** Data security
- — **easier to meet future data residency or audit requirements per school.** Regulatory compliance
- — **each school can have a completely different table structure based on their customizations, which is impossible in a shared schema model.** Schema freedom
- — **restore one school's data without touching any other school.** Backup & restore independence
- — **a heavy-query school does not slow down others.** Performance isolation

PostgreSQL is the chosen engine. It is the most mature relational database for complex transactional data (fees, payroll, attendance), has strong JSON support for semi-structured custom fields, supports row-level security, and has excellent managed offerings on all three major clouds: GCP Cloud SQL, AWS RDS, and Azure Database for PostgreSQL.

11.3 Redis — Per School

Each school also gets a dedicated Redis instance. Redis serves four critical roles:

Role	How Redis Is Used
Schema Migration Queue	When a school admin saves a workflow change or adds a field, the change is written to Redis as a job. The Schema Engine picks it up synchronously and executes the migration against PostgreSQL.
Real-Time Pub/Sub	Live dashboard updates, attendance broadcasts, fee payment confirmations, and notification dispatches flow through Redis pub/sub channels.
Session & Auth Cache	User sessions, role permission maps, and access control lists are cached so every API request does not hit PostgreSQL for auth checks.
Computed Data Cache	Expensive computed values — fee outstanding totals, attendance percentages, payroll summaries — are cached and invalidated on data change.

11.4 Cloud Provider Decision

Final cloud provider to be selected based on pricing analysis for the Indian market. All three are viable:

Provider	Pros for This Product
GCP	Strong managed PostgreSQL (Cloud SQL), excellent pricing in Mumbai region, Vertex AI integration for the AI agent. Strong choice if AI workloads are a priority.
AWS	Largest ecosystem, RDS PostgreSQL is battle-tested, ElastiCache for Redis, strong Indian region presence (Mumbai + Hyderabad). Best for enterprise reliability.
Azure	Best if target includes government schools — MeitY empanelled, strongest government cloud compliance in India. Slightly higher cost.

Recommendation

Run a 3-month cost comparison on equivalent workloads. Start with GCP for the build phase (free credits + Vertex AI for AI agent). Re-evaluate before production launch. Architecture is cloud-agnostic by design — switching providers requires only config changes, not code rewrites.

11.5 Automatic Provisioning Pipeline

When a school completes payment, the Provisioning Service runs fully automatically with zero human intervention. The school admin receives login credentials within minutes.

Step	What Happens
Step 1: Payment Confirmed	Billing service confirms payment. Triggers a provisioning job with school metadata (name, tier, region, admin email).
Step 2: Provision Database	Provisioning Service calls the cloud SDK to spin up a new PostgreSQL instance. Assigns a unique connection string stored in secrets manager.
Step 3: Seed Core Schema	The Core Schema Seeder runs all base migrations — creating all default tables (students, staff, fees, attendance, etc.) from our master schema template.
Step 4: Provision Redis	Spin up Redis instance for the school. Store connection details securely. Never exposed to school users.
Step 5: Provision Storage Bucket	Create dedicated file storage for documents, photos, and reports. Accessible only through our API, never directly.
Step 6: Register in Control Plane	Write school record to master registry DB: school ID, tier, resource pointers, provisioning timestamp, status = ACTIVE.
Step 7: Create Super Admin	Generate super admin credentials. Send secure welcome email with one-time password setup link.
Step 8: Monitoring Hooks	Register school resources in our monitoring dashboard. Set

	automated alerts for DB size thresholds, error rates, uptime.
--	---

School's Perspective

The school admin pays → receives a welcome email → clicks the link → sets their password → lands on the platform dashboard. They never see an IP address, a database name, a cloud console, or any infrastructure detail. Everything is behind the UI.

11.6 Live Schema Migration Engine

When a school admin adds a field, creates a new workflow node, or defines a new data entity on the canvas — the backend database updates in real time while the admin is watching. This is synchronous by design.

Stage	What Happens
1. UI Change Saved	Admin saves a canvas change. Frontend sends a Schema Change Event to the backend API.
2. Change Parsed	Schema Engine parses the change into a migration descriptor: { action: ADD_COLUMN, table: students, column: blood_group, type: VARCHAR(10) }.
3. Safety Check	Engine checks: Does the change conflict with existing data? Is the column name reserved? Will this break existing workflows? If unsafe, error returned to UI immediately.
4. Migration Executed	If safe, migration runs against the school's PostgreSQL instance synchronously. Simple changes (add column, new table) complete in milliseconds.
5. Meta-Schema Updated	The platform's internal meta-schema record for this school is updated. This is what the frontend reads to know what fields and tables exist.
6. Version Logged	Every migration is logged with timestamp, user, change descriptor, and rollback SQL. Full history for audit and recovery.
7. Confirmation to UI	Success response sent. Canvas unlocks. Admin sees their change reflected immediately.

Destructive changes (deleting a field or table with existing data) follow a soft-delete pattern: the column is hidden in the meta-schema but retained in the database until the admin explicitly confirms permanent deletion after reviewing a data impact report.

11.7 Security Model

- — **direct database access is never granted to any school user, role, or integration.** Schools access data only through our API layer
- — **never in code, never in environment variables, never exposed in logs.** All credentials in cloud secrets manager

- — **AES-256 at rest, TLS 1.3 in transit.** All data encrypted at rest and in transit
- — **every API query is scoped to the requesting school's tenant ID, enforced at the API layer before hitting the database.** API-level tenant scoping
- — **if a school ever needs raw data access for an audit, they submit a formal request. Our team reviews it, grants time-limited read-only access through a controlled interface, and logs the entire session.** Cloud access request flow
- — **point-in-time recovery retained for 30 days on Growth/Enterprise tiers, 7 days on Starter.** Automated daily backups

12. Open Questions & Next Steps

12.1 Product Decisions Pending

- Final product name selection from shortlist
- Mobile strategy: native app vs PWA for parent/student portals
- Offline capability: partial offline mode for attendance/fee collection in low-connectivity areas
- Multi-language UI: Hindi, Telugu, Tamil, Kannada, Marathi support timeline
- CBSE, ICSE, State Board specific compliance modules — which boards to support at launch
- Payment gateway integration for fee collection (Razorpay, PayU, Cashfree)

12.2 Business Decisions Pending

- Pricing — exact tier pricing in INR
- Go-to-market — direct sales vs channel partners (education consultants, school associations)
- Pilot school program — identify 2-3 schools for beta testing and feedback
- Company-assisted setup pricing and process
- Support model — chat support, phone support, dedicated account manager threshold

12.3 Technical Decisions Pending

- Schema-per-tenant vs row-level isolation for multi-tenancy
- How to handle schema rollback if a workflow/field is deleted that has existing data
- Canvas builder technology — evaluate React Flow, custom canvas, or hybrid approach
- AI agent context management — how much school data context to pass per query
- Voice agent language model fine-tuning for Indian school terminology
- Soft-delete vs hard-delete policy for removed schema elements with existing data
- Schema migration rollback strategy — automated or manual approval for destructive rollbacks

13. MVP Scope — v1.0 Definition

v1.0 is not a lite product. It is a complete, production-grade school management platform that ships with every major module, full drag-and-drop customization, live schema sync, a native mobile app, multi-branch support, full multi-language UI, and a built-in AI agent. Every school that sees it should be stunned.

v1.0 Philosophy

No shortcuts. No 'we'll add that in v2'. If a school needs it to run their operations end-to-end, it ships in v1.0. The product launches once — and it launches complete.

13.1 What Ships in v1.0

Platform Foundation

- Super admin account with full system access
- Drag-and-drop Role Builder canvas — create, edit, clone, delete roles with full permission mapping
- Drag-and-drop Workflow Builder canvas — visual node-based workflow design
- Live Schema Migration Engine — real-time backend sync on every canvas change
- Multi-branch architecture — manage multiple school branches under one account with branch-level isolation
- Full multi-language UI — English, Hindi, Telugu, Tamil, Kannada, Marathi from day one
- Custom UI component library — every component built in-house, no third-party kits
- Automatic cloud provisioning — school goes live within minutes of payment

Academic Management — Full Suite

- Admissions & Enrollment — application forms, document collection, offer letters, enrollment confirmation
- Student Records — complete profiles, academic history, medical info, family details
- Timetable & Scheduling — drag-and-drop period builder, teacher allocation, clash detection
- Attendance — daily and period-wise, biometric integration hooks, parent alerts
- Examination & Results — scheduling, marks entry, grade computation, report card generation, rank lists
- Curriculum Management — lesson plans, syllabus tracking, learning outcomes

Finance Management — Full Suite

- Fee Management — structure builder by class/category/installment, online & offline collection, receipts, defaulter tracking
- Budgeting — annual budget planning, department allocation, expense tracking, variance analysis

- Payroll & Salaries — computation, allowances, deductions, PF/ESI, TDS, payslips, bank transfer files
- Accounts & Ledger — income/expense ledger, day book, bank reconciliation, financial statements
- Vendor & Procurement — vendor management, purchase orders, invoice tracking, payment approvals
- Payment Gateway Integration — Razorpay + PayU for online fee collection

Human Resources — Full Suite

- Staff Profiles — employee records, qualifications, documents, contracts
- Recruitment — job postings, application tracking, interview scheduling, offer management
- Leave Management — types, approval workflows, balance tracking
- Performance Management — appraisal cycles, goal setting, review forms, increment workflows
- Training & Development — calendar, attendance, certifications, skill tracking

Operations Management — Full Suite

- Transport — route management, vehicle tracking, driver records, student bus assignment, GPS hook
- Hostel — room allocation, fee management, visitor log, warden dashboard
- Library — catalog, issue/return, fine management, digital resources
- Inventory & Assets — asset register, maintenance scheduling, depreciation, consumables stock
- Canteen — menu management, daily orders, billing
- Events & Activities — event calendar, participation tracking, co-curricular records

Communication & Portals

- Parent Portal — fee payment, attendance, results, teacher messaging, announcements
- Student Portal — homework, results, timetable, leave requests, digital content
- Internal Communication — staff announcements, circulars, noticeboard, messaging
- SMS + WhatsApp Integration — automated alerts for fees, attendance, results, events
- Native Mobile App (iOS + Android) — parent and student portals fully native, not just web wrapper

Education Content & Learning (Vidyut Learn)

- Digital book library — upload, organise, and assign digital content by class and subject
- Teacher Content Studio — rich lesson creation: text, images, video, audio, PDFs, quizzes
- Assignment Builder — MCQ + essay + file submission, auto-grading, AI bulk feedback
- Full Learning Calendar — assignments, exams, events, milestones in one synced view
- Live class scheduling with recording stored in lesson library
- Student Progress Dashboard — scores, time-on-task, weak topic identification
- AI-powered personalised revision suggestions per student

Device & Tablet Tier (Vidyut Device)

- Offline-first tablet app — full content access without internet
- Device Management Dashboard — battery, sync status, content push for all devices
- Content Sync Engine — teacher publishes, tablets receive automatically on next Wi-Fi
- Usage analytics per student per device
- Parental controls and school-curated content profile
- Hardware Partnership Programme — bulk tablet sourcing from vetted vendors
- Multi-device shared profiles for government schools

Compliance & Analytics

- Government Compliance — UDISE data, RTE compliance, state board reporting
- Analytics Dashboard — school-wide KPIs, attendance trends, fee analytics, academic performance
- Custom Report Builder — drag-and-drop reports from any data in the system

AI Agent — Full Integration

- Chat + voice interface available on every screen including the learning platform
- Full system configuration via conversation — roles, workflows, fee structures, timetables
- Teacher AI assistant — create assignments, generate quiz questions, draft lesson plans
- Student AI tutor — answer subject questions, explain concepts, suggest practice exercises
- Conversational data queries — any report, any filter, in plain language
- Proactive anomaly alerts — fee defaults, attendance drops, assignment non-submissions
- Per-role contextual help — onboards every user type from day one
- Indian English + Hindi voice support with regional accent handling

13.2 Build Complexity Honest Assessment

This is a large build. Below is an honest complexity rating for each major piece so engineering effort can be planned accurately.

Component	Complexity & Key Challenge
Live Schema Migration Engine	VERY HIGH — the most complex piece. Synchronous, transactional, rollback-safe schema changes per school in real time. Needs extensive testing for edge cases, data loss prevention, and concurrent user scenarios.
Drag-and-drop Canvas Builder	HIGH — a custom Figma-like node canvas for roles and workflows. Needs a graph rendering engine, state management for complex node relationships, and real-time preview of permission impacts.
Multi-language UI (6 languages)	HIGH — not just translation. RTL considerations, font rendering, dynamic string management, and ensuring every dynamically

	generated label (custom fields, custom roles) is also translatable.
Native Mobile App (iOS + Android)	HIGH — a full React Native app for parent and student portals. Push notifications, offline mode for attendance/fee viewing, biometric login.
AI Agent (chat + voice)	HIGH — context-aware per screen, per role, per school's custom schema. Voice STT/TTS with Indian accent support. Agent must understand each school's unique data model.
Automatic Cloud Provisioning	MEDIUM-HIGH — needs a robust, idempotent provisioning pipeline with failure recovery. Partial provisioning failures must be handled gracefully.
Multi-branch Architecture	MEDIUM-HIGH — branch-level data isolation within a school's database, branch-specific roles and permissions, cross-branch reporting for the owner.
Core Modules (Academic, Finance, HR, Ops)	MEDIUM — large surface area but well-understood domain. The main challenge is depth and Indian-specific requirements (TDS, PF/ESI, CBSE/ICSE/State board formats).
Payment Gateway Integration	MEDIUM — Razorpay and PayU both have mature APIs. Main complexity is reconciliation, refund handling, and failed payment workflows.
Custom Report Builder	MEDIUM — drag-and-drop report designer that queries any table in the school's dynamic schema. Needs a query builder that is safe against injection and schema-aware.

13.3 Suggested Build Sequence

Even without a timeline deadline, the build should follow a logical dependency order. Some components must exist before others can be built on top of them.

Phase	What Gets Built
Phase 1 — Foundation	Cloud provisioning pipeline. Control plane + master registry. Per-school PostgreSQL + Redis setup. Core schema seeder. Super admin auth. Basic dashboard shell.
Phase 2 — Schema Engine + Canvas	Live Schema Migration Engine. Role Builder canvas. Workflow Builder canvas. Permission enforcement layer. This phase is the hardest and unlocks everything else.
Phase 3 — Core Modules	Academic suite (admissions, attendance, timetable, exams). Finance suite (fees, payroll, accounts). HR suite. These are built on top of the dynamic schema from Phase 2.
Phase 4 — Operations + Comms	Transport, hostel, library, inventory, canteen, events. Parent and student portals (web). SMS + WhatsApp integration.
Phase 5 — Mobile + AI Agent	Native iOS + Android app. AI agent chat and voice integration. Agent training on school domain knowledge and per-school schema awareness.

Phase 6 — Polish + Scale	Multi-language UI rollout. Advanced analytics. Custom report builder. Government compliance modules. Performance tuning. Security audit. Beta with pilot schools.
Launch	Full production launch after Phase 6 sign-off with at least 3 pilot schools validating the complete product.

Key Insight on Sequencing

Phase 2 is the make-or-break phase. The Schema Engine and Canvas are what make this product unique — and they are also the riskiest to build. Everything in Phases 3-6 is built on top of them. Front-load the hardest engineering work early so the rest of the build flows naturally.

13.4 What Is Intentionally Deferred to v2.0

Even with a complete v1.0, a few items are better deferred to v2.0 to keep the build focused and avoid scope creep during development.

- **allow third-party developers to build modules on top of the platform. Powerful but complex governance challenge.** Marketplace / Plugin System
- **dropout risk prediction, fee default prediction, teacher performance forecasting using ML models trained on school data.** Advanced AI — predictive analytics
- **currency, tax, and compliance modules for Southeast Asia and Middle East markets.** International expansion features
- **anonymized cross-school performance comparisons (opt-in). Requires aggregating data across the platform.** School-to-school benchmarking
- **full offline capability with sync. Complex conflict resolution. v1.0 mobile app requires internet; offline mode in v2.0.** Offline-first mobile app
- **allow schools to brand the platform with their own domain and logo throughout. v1.0 uses the platform's own branding.** Custom domain + white-labeling

14. Tech Stack — Finalized

All technology choices are locked based on the latest stable versions as of 2026 and the product's specific requirements: Apple/Tesla-level UI quality, real-time schema sync, AI agent integration, full multi-language support, and a native mobile app.

14.1 Frontend — Web

Technology	Role & Rationale
React 19.2	Latest stable. React Compiler (auto-memoization, zero manual useMemo/useCallback), Server Components, <Activity> for instant tab switching on dashboards, View Transitions for seamless page navigation. Core framework for the entire web UI.
Next.js 16.1	Latest stable. Turbopack bundler (5-10x faster builds), Partial Pre-

	Rendering (PPR) for instant page loads, Cache Components for navigation speed, built-in React Compiler support. Framework shell for the web app.
TypeScript 5.x	Strict type safety across the entire codebase. Mandatory — no JavaScript in this project. Every API response, every schema event, every canvas action is fully typed.
Tailwind CSS (tokens only)	Not used for components — used only as a design token system for spacing, color variables, and typography scales. All actual components are hand-built. Zero pre-built component classes used.
Zustand	Lightweight state management for the canvas builder and global app state. Simpler than Redux, perfect for complex canvas node graphs and real-time schema state.
React Flow	Open-source graph engine for the drag-and-drop canvas builder (role hierarchy, workflow nodes). Completely custom-skinned — no default React Flow UI visible. The engine does the graph math, we own the visuals 100%.
Motion (Framer Motion v12+)	Declarative animation library for React. Powers all micro-interactions, page transitions, panel slides, modal animations, drag feedback. 39M weekly downloads, 85/100 health score. Used by Stripe, Notion, and Framer themselves.
GSAP (GreenSock)	Used alongside Motion for scroll-driven cinematic animations on the marketing site and onboarding flows — the Apple/Tesla-style reveal sequences. Industry standard for premium web animation. 26.7KB gzipped core.
React Native 0.83	Latest stable. iOS + Android mobile app for parent and student portals. Ships with React 19.2 features including <code><Activity></code> and <code>useEffectEvent</code> . New Architecture (Fabric renderer) for native-level performance.

14.2 Backend

Technology	Role & Rationale
Node.js LTS (v22+)	Runtime for the API layer. LTS channel for stability. Node.js v22 brings significant V8 performance improvements and native TypeScript support in upcoming versions.
Fastify v5.8	Latest stable. API framework. 2-3x faster than Express, native TypeScript, JSON Schema validation built-in, plugin architecture. Current main branch is v5. The right choice for high-throughput school API calls.
TypeScript (shared)	Monorepo shares TypeScript types between frontend and backend via a shared package. A schema change on the backend is immediately reflected in the frontend types — zero drift.
BullMQ + Redis	Job queue for schema migration events, background report generation, bulk notification dispatch, and payroll processing jobs. BullMQ is the modern successor to Bull, built on Redis Streams.

Prisma ORM	Type-safe database client for PostgreSQL. Auto-generates TypeScript types from the database schema. Used for the static/core tables. The dynamic meta-schema layer has its own raw SQL execution path.
Socket.io / WebSockets	Real-time bidirectional communication for live dashboard updates, attendance broadcasts, canvas collaboration (if multiple admins edit simultaneously), and notification push.

14.3 Database & Storage

Technology	Role & Rationale
PostgreSQL 16 (per school)	Primary data store. One fully isolated instance per school. Handles all transactional data — students, fees, attendance, payroll, HR. ACID compliant, row-level security, strong JSON support for flexible fields.
Redis 7 (per school)	Caching, session storage, BullMQ job queues, real-time pub/sub channels. Separate logical instance per school.
GCP Cloud Storage	File storage for documents, photos, report PDFs, payslips. Bucket per school. Never publicly accessible — all files served through signed URLs via the API.

14.4 AI & Voice

Technology	Role & Rationale
Claude API (Anthropic)	claude-sonnet model for the AI agent reasoning layer. Context-aware per screen, per role, per school's dynamic schema. Handles configuration conversations, data queries, workflow generation, and proactive alerts.
Sarvam AI	Indian-built STT/TTS platform. Handles voice input/output for the AI agent. Supports Indian English with regional accent handling, Hindi, and major Indian languages. Purpose-built for the Indian market — no other platform matches it for Indian voices.
Whisper (fallback)	OpenAI's Whisper as fallback STT for global English accuracy. Used when Sarvam AI confidence is below threshold.

14.5 Infrastructure & DevOps

Technology	Role & Rationale
GCP (Phase 1)	Google Cloud Platform for the build and early production phase. Cloud SQL (PostgreSQL), Memorystore (Redis), Cloud Run (API containers), Cloud Storage, Vertex AI (AI workloads). Mumbai region (asia-south1) for India-first performance.

Turborepo	Monorepo management. One repository contains: web app (Next.js), mobile app (React Native), API (Fastify), shared types package, shared UI tokens package, provisioning service, schema engine. Incremental builds — only rebuilds what changed.
Docker + Cloud Run	API services containerised with Docker. Deployed on GCP Cloud Run — serverless containers, auto-scale to zero when idle (cost saving), scales up instantly under load.
Terraform	Infrastructure-as-Code for repeatable, auditable cloud resource provisioning. The Provisioning Service uses the Terraform SDK under the hood to spin up per-school resources.
GitHub Actions	CI/CD pipeline. Automated testing, type checking, build verification, and deployment on every push to main. Zero manual deployments.
Sentry	Error tracking and performance monitoring for both web and mobile apps. Real-time alerts on production errors.

14.6 Authentication & Security

Technology	Role & Rationale
Clerk or Auth.js	Authentication provider for the web app. Handles email/password, magic links, 2FA, and session management. Integrates natively with Next.js 16 App Router. Decision: evaluate Clerk (managed, faster to build) vs Auth.js (self-hosted, more control).
GCP Secret Manager	All database credentials, API keys, Redis connection strings stored in Secret Manager. Never in environment variables or code. Rotated automatically.
Helmet.js + Rate Limiting	API security hardening on Fastify — HTTP headers, CORS, rate limiting per tenant, injection protection.

Version Summary

React 19.2 + Next.js 16.1 + React Native 0.83 + Fastify v5 + PostgreSQL 16 + Redis 7 + Turborepo + GCP + Claude API + Sarvam AI. Every version confirmed as latest stable as of 2026.

15. Frontend Architecture & Design Philosophy

The UI ambition is Apple and Tesla level — meaning every interaction feels intentional, every animation has purpose, every screen breathes. Not flashy. Not loud. Calm, fast, beautiful, and deeply functional. A school principal who has never used enterprise software should feel immediately at home.

Design Benchmark

Apple.com for product storytelling. Tesla's car dashboard UI for data-dense interfaces. Linear.app for

SaaS product UI clarity. Notion for information architecture. None of these are Indian school ERP competitors — they are the bar we are setting.

15.1 Design System — Built In-House

Every single component is hand-crafted. No Material UI, no Radix, no Shadcn, no Ant Design. This is non-negotiable — third-party component libraries always look generic and impose constraints that make Apple-level UI impossible.

System	Approach
Design Tokens	All spacing, color, typography, border radius, shadow, and motion values defined as CSS custom properties. One source of truth. Light and dark mode built in from day one.
Typography Scale	Custom type scale with a premium typeface (likely Inter or a custom licensed font). Clear hierarchy: display, heading, subheading, body, caption, label. Every text element maps to exactly one style.
Color System	A primary palette (brand blue-grey spectrum), a semantic palette (success, warning, error, info), and a neutral palette. All colors pass WCAG AA contrast. Dark mode is not an afterthought — it's designed simultaneously with light.
Motion System	All animations defined as tokens: duration-fast (100ms), duration-normal (200ms), duration-slow (350ms), easing-spring, easing-smooth. No ad-hoc animation values anywhere. Motion used for utility, not decoration.
Component Library	Buttons, inputs, selects, modals, drawers, tooltips, tables, cards, badges, tabs, breadcrumbs, calendars, data grids — all custom. Each component has: default, hover, active, focus, disabled, error states. Fully accessible (ARIA labels, keyboard navigation).
Icon System	Custom SVG icon set or a single premium icon library (Phosphor Icons — most complete, consistent, MIT licensed). Never mix icon libraries.

15.2 Animation Strategy — Apple/Tesla Standard

The goal is motion that makes the product feel alive without ever feeling busy. Every animation serves a purpose: orient the user, confirm an action, or reveal information progressively.

Surface	Animation Approach
Page Transitions	Next.js 16 View Transitions API + Motion. Pages slide, fade, or morph depending on context. Navigation feels instant — not like loading, like moving within a space.
Micro-interactions	Every button press, every toggle, every form field has a subtle

	physical response. Scale on press, color shift on hover, spring bounce on modal open. Motion handles all of this declaratively.
Canvas Animations	The drag-and-drop role and workflow canvas uses smooth physics-based drag (Motion's drag API), spring-snapping to grid, and silky connection line rendering via React Flow.
Data Loading	Skeleton screens that match the exact shape of the content they replace — never generic grey boxes. Skeleton animation is a subtle shimmer, not a pulse.
Scroll Reveals	GSAP ScrollTrigger powers the marketing/onboarding pages. Content reveals as you scroll — the same technique Apple uses on product pages. Each section choreographed, not random.
Dashboard Transitions	Numbers count up on load. Chart bars grow in. Module cards stagger-fade in. The dashboard feels like it's waking up when you arrive. All performant — GPU-accelerated transforms only.

15.3 Canvas Builder UI

The role and workflow canvas is the most complex UI surface in the entire product. It must feel like using a tool built by a design company, not an ERP vendor.

- Infinite canvas with smooth pan and zoom (like Figma — pinch-to-zoom on trackpad, scroll-to-zoom on mouse)
- Nodes snap to an invisible grid — clean alignment without manual effort
- Connection lines between nodes are smooth bezier curves that reroute automatically as nodes move
- Right-click context menu on any node for quick actions (edit, duplicate, delete, add connected node)
- Mini-map in corner for navigation on large canvases
- Undo/redo with full history — Ctrl+Z works like Figma
- Real-time preview panel alongside canvas — changes reflect live without saving
- AI agent accessible directly from canvas — describe what you want, agent builds the nodes

15.4 Mobile App Design

The React Native app for parents and students must feel like a premium consumer app, not a school portal. Reference: Apple's own apps, Zepto, CRED — apps Indians already consider beautiful.

- Bottom tab navigation (not hamburger menus) — thumb-friendly, always visible
- Haptic feedback on key actions — fee payment confirmation, attendance marked, result published
- Biometric login (Face ID / fingerprint) from day one — no parent wants to type a password
- Push notifications with rich media — not just text alerts, but contextual cards
- Offline viewing for key data — timetable, attendance history, fee receipts viewable without internet
- Pull-to-refresh with custom animation — not the default grey spinner

16. Pricing Strategy

Pricing is set at a 3x minimum markup over true cost-to-serve, anchored below existing Indian school ERP alternatives to create undeniable value, while reflecting the dramatically superior product. Schools switching from Entab or Fedena actually save money while getting a better product.

16.1 Subscription Tiers — Monthly (billed annually at 15% discount)

Tier & Price	What's Included
Starter — ₹14,999/month	Up to 500 students. All core modules. Single branch. AI agent (chat + voice). Parent + student web portal. Mobile app access. Email + chat support. Annual = ₹1,52,990/year.
Growth — ₹34,999/month	Up to 2,000 students. Everything in Starter + multi-branch (up to 3 branches) + advanced HR + payroll + transport + hostel + library + WhatsApp integration + priority support. Annual = ₹3,56,990/year.
Enterprise — ₹74,999/month	Unlimited students, unlimited branches. Everything in Growth + custom module requests + dedicated account manager + company-assisted onboarding included + SLA guarantee + compliance reporting. Annual = ₹7,64,990/year.

16.2 Cloud Cost vs Revenue (Monthly)

Scenario	Numbers
Starter school cost to serve	~₹6,300/month (PostgreSQL ₹2,100 + Redis ₹1,350 + Storage ₹200 + API ₹420 + Overhead ₹2,230)
Starter margin	₹14,999 revenue – ₹6,300 cost = ₹8,699 gross profit. 58% gross margin.
Growth school cost to serve	~₹13,000/month (larger DB instance + more storage + higher AI usage + SMS volume)
Growth margin	₹34,999 revenue – ₹13,000 cost = ₹21,999 gross profit. 63% gross margin.
Enterprise school cost to serve	~₹24,300/month (large DB, high storage, heavy AI, high volume comms)
Enterprise margin	₹74,999 revenue – ₹24,300 cost = ₹50,699 gross profit. 68% gross margin.

16.3 ARR Projections

Milestone	Revenue Estimate
50 schools (Year 1 target)	Mixed tier (30 Starter, 15 Growth, 5 Enterprise): ₹54L + ₹63L + ₹45L = ₹1.62Cr ARR
150 schools (Year 2 target)	Mixed tier (70 Starter, 55 Growth, 25 Enterprise): ₹1.26Cr + ₹2.31Cr + ₹2.25Cr = ₹5.82Cr ARR
300 schools (Year 3 target)	Mixed tier: approximately ₹12–15Cr ARR depending on tier distribution
500 schools (Scale target)	Mixed tier: approximately ₹20–25Cr ARR. Unit economics are strong — each new school is primarily margin.

16.4 Value vs Competitors

Comparison	Analysis
Entab / CampusCare	₹60,000–₹3,00,000/year depending on school size. Rigid software. No AI. No drag-and-drop. No real-time schema sync. Same or higher price as our platform with far less capability.
Fedena	₹35,000–₹1,60,000/year. Open-source base, dated UI, requires technical team to customize. No AI agent.
Our Starter tier vs Entab small school	₹1,52,990/year vs ₹60,000–₹80,000/year for Entab. We are priced higher — and justified because we deliver dramatically more. The school is not buying a cheaper product; they are buying a generational upgrade.
Cost per student (Starter, 300 students)	₹50/student/month. Less than a school diary. Less than a photocopy of a timetable. The value proposition per student is undeniable.

Pricing Philosophy

We are not competing on price. We are competing on value. A school that sees this product running in a competitor school will not ask 'is it cheaper than what we have?' — they will ask 'how do we get this?' Price accordingly. Never discount the core product — offer extended trials instead.

16.5 Additional Revenue

- **₹25,000 one-time fee for white-glove setup. Starter schools can self-setup; Growth/Enterprise is offered this by default.** Company-assisted onboarding
- **₹50,000–₹2,00,000 per bespoke module depending on complexity. Enterprise clients only.** Custom module development
- **included in subscription. Not a separate line item — competitor AMCs are a hidden cost we fold into the subscription.** Annual maintenance contract (AMC)

- — ₹5,000–₹15,000 per session for staff training workshops. Scalable as recorded online courses later. Training programs

17. Innovation Modules — Platform Differentiators

The following 12 modules go beyond standard school ERP functionality. Each solves a genuine Indian school problem that no competitor currently addresses. They are the features that will make schools say 'no other platform does this' and create category leadership over the next 5 years.

Prioritisation Note

Priority 1 (v1.5 — ship next after v1.0): Transport Live Tracking, Fee Recovery Engine, Blockchain Marksheets, Vendor Marketplace. Priority 2 (v2.0): Teacher Performance Intelligence, Student Mental Health, Regulatory Compliance Autopilot, Canteen Digital Wallet. Priority 3 (v2.5+): Alumni Network, Parent Trust Score, Substitute Teacher Automation, Competitor Benchmarking.

17.1 Smart Fee Recovery Engine

Indian schools lose 15–30% of annual fee revenue to defaulters and delayed payments. This module does not just flag defaulters — it predicts which families are likely to default before they do, and automates a graduated recovery response.

Feature	Detail
Prediction Model	Machine learning on payment history patterns, sibling count, class, seasonal trends, and historical default behaviour. Flags families with >70% predicted default probability 30 days before due date.
Graduated Response	AI-triggered escalation: Day -30 → friendly WhatsApp reminder. Day -7 → personal call request. Day 0 → overdue notice. Day +7 → fee restructuring offer (EMI plan). Day +30 → legal notice template auto-drafted.
EMI Restructuring	School admin can approve a payment plan in one click. System breaks outstanding fees into installments, updates fee schedule, tracks each installment separately.
Recovery Analytics	Dashboard showing: predicted defaulters this month, amount at risk, recovery rate trend, which communication channel has highest response rate.
Payment Gateway Hook	Razorpay payment links generated per family, sent via WhatsApp. Parent pays in one tap — no portal login required.

Revenue Impact for Schools

A school with ₹1 crore annual fee revenue losing 20% to defaults recovers ₹15–18 lakh annually with this module — making the platform subscription cost essentially zero on an ROI basis.

17.2 Parent Trust Score

A behavioural scoring system for parents that gives school management objective data for admissions prioritisation and engagement targeting.

Feature	Detail
Score Components	Fee payment consistency (40%), attendance at PTMs (20%), response rate to school communications (20%), volunteer participation (10%), sibling enrolment history (10%).
Score Range	0–100. Green (75+): highly engaged. Amber (50–74): moderate engagement. Red (<50): needs outreach.
Use Cases	Admissions: when two applicants are equal, prefer families with higher trust scores. Fee planning: flag high-risk families for proactive payment plans. Engagement: auto-assign school counsellor outreach to low-score families.
Privacy Controls	Scores are internal only — never shown to parents. Super admin can configure which factors are included. Can be disabled entirely if school prefers not to use it.

17.3 Blockchain-Verified Digital Marksheets

Every student result document gets a tamper-proof QR code backed by a public blockchain record. Solves India's widespread fake certificate problem and makes the platform's marksheets the most trusted in the country.

Feature	Detail
How It Works	When a result is finalised and approved by the principal, a SHA-256 hash of the student's result data is written to a public blockchain (Polygon — low cost, fast finality). A QR code embedding the transaction ID is printed on the marksheet.
Verification	Anyone (college admissions, employer, embassy) scans the QR code → sees the exact result data on a public verification page → confirms it matches the physical document. No login, no calling the school.
Tamper Detection	Any change to the marksheet after blockchain commit is immediately detectable — the hash no longer matches. Fraudulent certificates are caught instantly.
Bulk Issuance	End-of-year result processing issues all certificates in one batch. Each certificate gets its own blockchain record. Process takes minutes regardless of school size.
Cost	Polygon transaction cost: ~₹0.01 per certificate. Essentially free. No per-student licensing fee.

17.4 Teacher Performance Intelligence

The only Indian school ERP module that gives principals objective, data-driven insight into teacher effectiveness — without making teachers feel surveilled.

Feature	Detail
Data Sources	Student attendance rates by teacher (proxy for classroom engagement), exam score trends per class per teacher year-over-year, parent feedback scores (optional, anonymous), syllabus completion rate, peer observation logs entered by HODs.
Performance Dashboard	Principal sees each teacher's composite score across all dimensions. Drill down into any metric. Compare teacher to department average and school average.
Appraisal Integration	Performance data auto-populates the annual appraisal form. Teacher sees their own data during self-review. Manager adds qualitative notes. Increment recommendation generated as a starting point.
Development Paths	AI suggests training interventions based on weak areas. Low exam result trend → suggests pedagogy workshop. Low attendance rate → suggests classroom management resource.
Privacy	Individual teacher scores visible only to HOD, principal, and HR. Never visible to other teachers or parents.

17.5 Transport Live Tracking

The highest-demand parent feature in Indian schools. Not just route management — a live map showing exactly where the school van is right now, with accurate ETA at each stop.

Feature	Detail
Driver App	Simple React Native app for drivers. One button to start route, GPS shares location automatically. No complex setup — driver just presses Start and drives.
Parent App	Live map in the parent mobile app showing van position. ETA calculated dynamically based on current location and traffic. Push notification when van is 10 minutes from stop.
School Dashboard	Operations manager sees all vans on one map. Flag late vans. Contact driver in one tap. View route completion status.
Route Optimisation	AI suggests optimised routes based on student home locations and historical traffic patterns. Reduces fuel cost and travel time.
Incident Alerts	If van deviates significantly from planned route or stops unexpectedly for >5 minutes, automatic alert fires to operations manager and parents.
Attendance Sync	Student marked as 'in transit' when they board the van (via driver

	app tap or NFC card). Marked 'arrived' when van reaches school. Attendance records automatically.
--	---

17.6 Canteen Digital Wallet

A closed-loop digital payment system inside the school. No cash, no change, no lost money. Parents control spending, students transact instantly, canteen operators get analytics.

Feature	Detail
Parent Wallet Top-Up	Parent loads wallet via the mobile app using Razorpay. Set daily spend limits per child. Get real-time notification of every purchase. View full transaction history.
Student Transaction	Student taps NFC card or shows QR code at canteen counter. Cashier scans, amount deducted instantly. Receipt sent to parent WhatsApp.
Canteen Analytics	Most popular items, peak hours, daily revenue, waste tracking. Canteen operator sees everything needed to manage inventory and staffing efficiently.
Nutrition Tracking	Each menu item tagged with nutritional category. Parent app shows weekly nutrition summary for their child's purchases. School can flag nutritionally poor purchase patterns.
Low Balance Alert	Automatic alert to parent when child's wallet balance falls below threshold (configurable). Top-up in one tap from the alert notification.

17.7 Substitute Teacher Automation

Eliminates the most stressful 30 minutes of a principal's morning — finding a substitute teacher when someone calls in sick.

Feature	Detail
Absence Trigger	Teacher marks themselves absent via mobile app by 7am (or admin marks them). System immediately begins finding a substitute.
Smart Matching	System checks: which teachers are free during the absent teacher's periods, which are qualified to teach the subject, which are in the same or nearby department. Ranked by availability and proximity.
One-Tap Accept	Top 3 substitute candidates get a push notification with the request. First to accept gets the assignment. Others get auto-cancelled notifications.
Auto-Update	Timetable updates automatically. Students and parents notified of the substitute. HOD gets a summary. All in under 5 minutes, zero

	manual calls.
Substitute Log	Full record of who substituted whom, how often, and for which subjects. Feeds into performance and workload tracking.

17.8 Student Mental Health Module

A confidential, counsellor-facing tool for proactive student wellbeing tracking. Addresses India's growing student mental health crisis with data-driven early intervention.

Feature	Detail
AI Concern Flags	System monitors: attendance drop >20% over 2 weeks, grade drop >15% in any subject, behavioral incident frequency increase, teacher concern flags. When 2+ signals appear simultaneously, counsellor is alerted to check in proactively.
Counsellor Dashboard	Private to counsellor only — not visible to teachers, principal, or parents. Shows students with active concern flags, session history, and intervention notes.
Session Notes	Counsellor logs check-in sessions. Notes encrypted and accessible only to the counsellor. Aggregate anonymised trends visible to principal (% of students seen, not individual names).
Referral Workflow	Counsellor can flag a student for external professional support. System generates a referral letter template and logs the recommendation. Parent is notified via a sensitive communication template.
Privacy Guarantee	Zero data visible to parents, teachers, or other students. Mental health records excluded from all standard reports and data exports. Separate access control layer.

17.9 Regulatory Compliance Autopilot

Schools spend dozens of hours annually manually compiling UDISE returns, RTE compliance reports, and state board submissions. This module eliminates that work entirely.

Feature	Detail
UDISE Auto-Fill	All UDISE data fields mapped to live school data. Student counts, teacher qualifications, infrastructure details, enrolment by category — auto-populated from the database. One-click export in the exact UDISE XML/CSV format.
RTE Compliance	Tracks 25% EWS/DG seat allocation, free uniform and textbook distribution records, age-appropriate admission compliance. Generates the annual RTE compliance certificate automatically.
State Board Reports	Configurable per board (CBSE, ICSE, state boards). Common

	annual returns auto-populated. School selects their board, system generates the correct format.
Submission Calendar	All regulatory deadlines tracked automatically. Reminder notifications to principal 30 days, 7 days, and 1 day before each deadline. Never miss a submission again.
Audit Ready	One-click generation of an audit-ready compliance package — all required documents, properly formatted, with a cover sheet listing every included document. Ready to hand to an inspector.

17.10 Alumni Network & Lifecycle Management

Schools lose all relationship with students the day they graduate. This module turns alumni into a long-term asset — for reputation, mentorship, and revenue.

Feature	Detail
Alumni Directory	Graduates automatically added to alumni directory on passing out. Can update their own profile: current institution, career, location. School can search and filter by batch, career path, location.
Job Placement Tracking	Alumni update employment status. School tracks: top universities attended by alumni, industry distribution, notable achievements. Builds genuine proof of outcomes for admissions marketing.
Mentorship Programme	Working alumni paired with current students for career guidance. Student applies for mentor in their area of interest. System suggests matches. Mentorship sessions logged.
Donation Campaigns	School creates fundraising campaigns for infrastructure, scholarships, or events. Alumni notified via email and app. Online donation via Razorpay. Receipt auto-generated. Tax exemption letter if school has 80G registration.
Alumni Events	Reunion management, homecoming events, batch WhatsApp group auto-creation. Alumni-only event registrations and ticketing.

17.11 Vendor Marketplace

Turns the platform from a cost center into a genuine money-saver. Aggregated purchasing power across hundreds of schools unlocks bulk pricing that individual schools could never negotiate alone.

Feature	Detail
Procurement Categories	Textbooks and workbooks, uniforms and sportswear, stationery and art supplies, lab equipment and chemicals, sports equipment, cleaning and hygiene supplies, furniture and fixtures, electronics.
How It Works	Platform aggregates demand across all member schools → negotiates bulk pricing with verified vendors → schools order at

	negotiated rates → vendor delivers directly to school → platform takes 3–5% commission.
Vendor Verification	All vendors undergo background check, GST verification, quality audit, and sample review before listing. Rating system after each delivery. Poor vendors removed.
Order Management	School creates purchase orders through the platform. Approval workflow (HOD → accountant → principal). PO sent to vendor automatically. Delivery tracked. Invoice matched to PO before payment released.
Savings Dashboard	School sees their cumulative savings vs market price since joining the marketplace. Typical saving: 15–25% on category spend. This is the platform paying for itself.
Network Effect	More schools → more purchasing volume → better vendor pricing → more value per school → more schools join. Classic marketplace flywheel.

17.12 Anonymous Competitor Benchmarking

Schools have no objective way to know how they compare to peers. This module gives them honest, anonymised benchmarking data that drives improvement — and creates a powerful network effect for the platform.

Feature	Detail
Opt-In Model	Schools choose to participate. Participating schools access benchmarking data. Non-participating schools see no data. Incentive to share is strong — you must give to receive.
Benchmark Dimensions	Fee collection rate (% collected vs billed), student attendance rate, exam pass percentage by subject, staff retention rate, parent app engagement rate, NPS score (if running parent surveys).
Peer Groups	Benchmarked against similar schools only: same tier (by student count), same city or region, same board affiliation. No comparing a 200-student school to a 3000-student school.
Insight Reports	Monthly report: 'Your fee collection rate is 78% vs peer average of 84%. Schools in the top quartile use automated payment reminders — activate this feature.' Actionable, not just informational.
Platform Value	Every new school that joins improves benchmarking quality for all existing schools. This is the most powerful network effect in the product — data gets more valuable as the platform grows.

Combined Impact of Innovation Modules

Transport tracking reduces parent anxiety. Fee recovery improves school revenue. Blockchain marksheets eliminate fraud. Vendor marketplace pays for the subscription. Benchmarking drives school improvement. Mental health module protects students. Together these 12 modules don't just

make the platform better — they make switching away from it unthinkable. The switching cost grows with every module a school activates.

Document Status

This document is the complete product brainstorm and vision reference: working name (Vidyut — rename anytime), product vision, naming research (Gurukrama.com confirmed available plus 5 new Sanskrit candidates), target market, business model with 5 tiers including Vidyut Learn and Vidyut Device add-ons, all 30+ core modules including full Education Content & Learning platform and Tablet/Device tier with hardware partnership programme, AI agent design with teacher and student assistance, canvas builder, full backend architecture, tech stack (latest 2026 versions locked), frontend design philosophy, MVP scope with build phases, competitive landscape, complete pricing strategy with margin analysis and ARR projections, and 12 innovation modules that create category leadership.

All technical design, CI/CD, security, API contracts, design system, AIIS, owner portal, cloud strategy, and developer workflow are documented in the companion Operations & Design Reference document.